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Practical Natural Language Processing

A Comprehensive Guide to Building
Real-World NLP Systems



Sowmya Vajjala,
Bodhisattwa Majumder,
Anuj Gupta & Harshit Surana

Praise for *Practical Natural Language Processing*

Practical NLP focuses squarely on an overlooked demographic: the practitioners and business leaders in industry! While many great books focus on ML's algorithmic fundamentals, this book exposes the anatomy of real-world systems: from e-commerce applications to virtual assistants. Painting a realistic picture of modern production systems, the book teaches not only deep learning, but also the heuristics and patchwork pipelines that define the (actual) state of the art for deployed NLP systems. The authors zoom out, teaching problem formulation, and aren't afraid to zoom in on the grimy details, including handling messy data and sustaining live systems. This book will prove invaluable to industry professionals keen to build and deploy NLP in the wild.

—Zachary Lipton, Assistant Professor, Carnegie Mellon University, Scientist at Amazon AI,
Author of *Dive into Deep Learning*

This book does a great job bridging the gap between natural language processing (NLP) research and practical applications. From healthcare to e-commerce and finance, it covers many of the most sought-after domains where NLP is being put to use and walks through core tasks in a clear and understandable manner. Overall, the book is a great manual on how to get the most out of current NLP in your industry.

—Sebastian Ruder, Research Scientist, Google DeepMind

There are two kinds of computer science books on the market: academic textbooks that give you a deep understanding of a domain but can be difficult to access for a non-academic, and “cookbooks” that outline solutions to very specific problems without providing the technical foundations that would allow the reader to generalize the recipes. This book offers the best of both worlds: it is thorough yet accessible. It provides the reader with a solid foundation in natural-language processing. . . . If you would like to go from zero to one in NLP, this book is for you!

—Marc Najork, Research Engineering Director, Google AI,
ACM & IEEE Fellow

There are text books or research papers or books on programming tips, but not a book that tells us how to build an end-to-end NLP system from scratch. I am happy to see this book on practical NLP, which fills this much needed gap. The authors have meticulously, thoughtfully and lucidly covered each and every aspect of NLP that one has to be aware of while building large scale practical systems; at the same time, this book has also managed to cover a large number of examples and varied application areas and verticals. This book is a must for all aspiring NLP engineers, entrepreneurs who want to build companies around language technologies, and also academic researchers who would like to see their inventions reach the real users.

—Monojit, Principal Researcher, Microsoft Research India, Adjunct Faculty at IIIT Hyderabad,
Ashoka University, IIT Kharagpur

This book bridges the gap between theory and practice by explaining the underlying concepts while keeping in mind varied real-world deployments across different business verticals. There is much hard-fought practical advice from the trenches whether it is about tweaking parameters of open source

libraries, setting up data pipelines for building models, or optimizing for fast inference. A must-read for engineers building NLP applications.

—Vinayak Hegde, CTO-in-Residence, Microsoft For Startups

This book shows how to put NLP to practice. It bridges the gap between NLP theory and practical engineering. The authors achieved a rare feat by simplifying the esoteric art of design and architecture of production quality machine learning systems.

I wish I had access to this book early on in my professional career and evaded the mistakes I made along the way. . . . I am deeply convinced that this book is an essential read for anybody aiming to develop involved in developing a robust, high-performing NLP system.

—Siddharth Sharma, ML Engineer, Facebook

I feel this is not only an essential book for NLP practitioners, it is also a valuable reference for the research community to understand the problem spaces in real-world applications. I very much appreciate this book and wish this could be a long-term project with up-to-date NLP application trending!

—Mengting Wan, Data Scientist (ML&NLP) at Airbnb, Microsoft Research Fellow

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Your File Format APIs

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Dedication

This book is dedicated to our respective advisors: Detmar Meurers, Julian McAuley, Kannan Srinathan, and Luis von Ahn.

Foreword

The field of natural language processing (NLP) has undergone a dramatic shift in recent years, both in terms of methodology and in terms of the applications supported. Methodological advances have

ranged from new ways of representing documents to new techniques for language synthesis. With these have come new applications ranging from open-ended conversational systems to techniques that use natural language for model interpretability. Finally, these advances have seen NLP gain a foothold in related areas, such as computer vision and recommender systems, some of which my lab is working on with support from Amazon, Samsung, and the National Science Foundation.

As NLP is expanding into these exciting new areas, so too has the audience of practitioners wanting to make use of NLP techniques. In the Data Science course (CSE 258) that I take at the University of California–San Diego, which is often the most attended in the computer science department, I see that more and more students are doing their projects on NLP-based topics. NLP is rapidly becoming a necessary skill required by engineers, product managers, scientists, students, and enthusiasts wishing to build applications on top of natural language data. On one hand, new tools and libraries for NLP and machine learning have made natural language modeling more accessible than ever. But on the other hand, resources for learning NLP must target this ever-growing and diverse audience. This is especially true for organizations that have recently adopted NLP or for students working with natural language data for the first time.

It has been my pleasure over the last few years to collaborate with Bodhisattwa Majumder on exciting new applications in NLP and dialog, so I was thrilled to hear about his efforts (along with Sowmya Vajjala, Anuj Gupta, and Harshit Surana) to write a book on NLP. They have a wide experience in scaling NLP including at early-stage startups, the MIT Media Lab, Microsoft Research, and Google AI.

I am excited by the end-to-end approach taken in their book, which will make it useful for a range of scenarios and will help readers to work with the labyrinth of possible options while building NLP applications. I am especially thrilled about the emphasis on modern NLP applications such as chatbots, as well as the focus on interdisciplinary topics such as ecommerce and retail. These topics will be especially useful for industry leaders and researchers, and are critical subjects that have been given only limited coverage in existing textbooks. This book is ideal both as a first resource to discover the field of natural language processing and a guide for seasoned practitioners looking to discover the latest developments in this exciting area.

Julian McAuley

Professor of Computer Science and Engineering

University of California, San Diego

Preface

Natural language processing (NLP) is a field at the intersection of computer science, artificial intelligence, and linguistics. It concerns building systems that can process and understand human language. Since its inception in the 1950s and until very recently, NLP has primarily been the domain of academia and research labs, requiring long formal education and training. The past decade's

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breakthroughs have resulted in NLP being increasingly used in a range of diverse domains such as retail, healthcare, finance, law, marketing, human resources, and many more. There are a range of driving forces for these developments:

- Widely available and easy-to-use NLP tools, techniques, and APIs are now all-pervading in the industry. There has never been a better time to build quick NLP solutions.
- Development of more interpretable and generalized approaches has improved the baseline performance for even complex NLP tasks, such as open-domain conversational tasks and question answering, which were not practically feasible before.
- More and more organizations, including Google, Microsoft, and Amazon, are investing heavily in more interactive consumer products, where language is used as the primary medium of communication.
- Increased availability of useful open source datasets, along with standard benchmarks on them, has acted as a catalyst in this revolution, as opposed to being impeded by proprietary datasets only available to limited organizations and individuals.
- The viability of NLP has moved beyond English or other major languages. Datasets and language-specific models are being created for the less-frequently digitized languages too. A fruitful product that came out of this effort was a near-perfect automatic machine translation tool available to all individuals with a smartphone.

With this rapidly expanding usage, a growing proportion of the workforce that builds these NLP systems is grappling with limited experience and theoretical knowledge about the topic. This book addresses this need from an applied perspective. Our book aims to guide the readers to build, iterate, and scale NLP systems in a business setting, and to tailor them for various industry verticals.

Why We Wrote This Book

There are many popular books on NLP available. While some of these serve as textbooks, focusing on theoretical aspects, some others aim to introduce NLP concepts through a lot of code examples. There are a few others that focus on specific NLP or machine learning libraries and provide “how-to” guides on solving different NLP problems using the libraries. So, why do we need another book on NLP?

We have been building and scaling NLP solutions for over a decade at leading universities and technology companies. While mentoring colleagues and other engineers, we noticed a gap between NLP practice in the industry and the NLP skill sets of new engineers and those who are just starting with NLP in particular. We started understanding these gaps even better during NLP workshops we were conducting for industry professionals, where we noticed that business and engineering leaders also have these gaps.

Most online courses and books tackle NLP problems using toy use cases and popular (often large, clean, and well-defined) datasets. While this imparts the general methods of NLP, we believe it does not provide enough of a foundation to tackle new problems and develop specific solutions in the real world. Commonly encountered problems while building real-world applications, such as data collection, working with noisy data and signals, incremental development of solutions, and issues involved in deploying the solutions as a part of a larger application, are not dealt with by existing resources, to the best of our knowledge. We also saw that best practices to develop NLP systems were missing in most scenarios. We felt a book was needed to bridge this gap, and that is how this book was born!

The Philosophy

We want to provide a holistic and practical perspective that enables the reader to successfully build real-world NLP solutions embedded in larger product setups. Thus, most chapters are accompanied by code walkthroughs in the associated Git repository. The book is also supplemented with extensive references for readers who want to delve deeper. Throughout the book, we start with a simple solution and incrementally build more complex solutions by taking a minimum viable product (MVP) approach, as commonly found in industry practice. We also give tips based on our experience and learnings. Where possible, each chapter is accompanied by a discussion on the state-of-the-art in that topic. Most chapters conclude with a case study of real-world use cases.

Consider the task of building a chatbot or text classification system at your organization. In the beginning there may be little or no data to work with. At this point, a basic solution using rule-based systems or traditional machine learning will be apt. As you accumulate more data, more sophisticated NLP techniques (which are often data intensive) can be used, including deep learning. At each step of this journey there are dozens of alternative approaches one can take. This book will help you navigate this maze of options.

Scope

This book provides a comprehensive view on building real-world NLP applications. We will cover the complete lifecycle of a typical NLP project—from data collection to deploying and monitoring the model. Some of these steps are applicable to any ML pipeline, while some are very specific to NLP. We also introduce task-specific case studies and domain-specific guides to build an NLP system from scratch. We specifically cover a gamut of tasks ranging from text classification to question answering to information extraction and dialog systems. Similarly, we provide recipes to apply these tasks in domains ranging from e-commerce to healthcare, social media, and finance. Owing to the depth and breadth of the topics and scenarios we cover, we will not go step by step explaining the code and all the concepts. For details of the implementation, we have provided detailed source code notebooks. The code snippets in this book cover the core logic and often skip introductory steps like setting up a library

or importing a package as they are covered in the associated notebooks. To cover the wide range of concepts we have given more than 450 extensive references to delve deeper into these topics. This book will be a day-to-day cookbook giving you a pragmatic view while building any NLP system, as well as be a stepping stone to broaden the application of NLP into your domain.

Who Should Read This Book

This book is for anyone involved in building NLP applications for real-world use cases. This includes software developers and testers, machine learning engineers, data engineers, MLOps engineers, NLP engineers, data scientists, product managers, people managers, VPs, CXOs, and startup founders. This also includes those involved in data creation and annotation processes—in short anyone and everyone who is involved in any way in building NLP systems in industry. While not all chapters are useful for people with all roles, we tried to give lucid explanations using less technical jargon and more intuitive understanding wherever possible. We believe there is something in every chapter for all potential readers interested in getting a holistic perspective about building NLP applications.

Some chapters or sections can be understood without much coding experience and code bits can be skipped as needed. For example, the first two sections in Chapter 1 and Chapter 9, or the sections “The Data Science Process” and “Making AI succeed in your organization” in Chapter 11 can be understood without any coding experience by all groups of readers. As you progress through the book, you will find more such sections in all chapters. However, to extract maximum benefit from this book, its notebooks, and references, we expect the reader to have the following background:

- Intermediate proficiency in Python programming. For example, understanding Python features such as list comprehension, writing functions and classes, and using existing libraries.
- Familiarity with various aspects of the software development life cycle (SDLC) such as design, development, testing, DevOps, etc.
- Basics of machine learning, including familiarity with commonly used machine learning algorithms such as logistic regression and decision trees and the ability to use them in Python with existing libraries such as scikit-learn.
- Basic knowledge of NLP is useful but not mandatory. Having an idea of tasks such as text classification and named entity recognition is also helpful.

What You Will Learn

Our primary audience is comprised of engineers and scientists involved in building real-world NLP systems for different verticals. Some of the common job titles are: Software Engineer, NLP Engineer, ML Engineer, and Data Scientist. The book may also be helpful for product managers and engineering leaders. However, it may not be as helpful for those pursuing cutting-edge research in NLP because we

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do not cover in-depth theoretical and technical details related to NLP concepts. With this book, you will:

- Understand the wide spectrum of problem statements, tasks, and solution approaches within NLP.
- Gain experience in implementing and evaluating different NLP applications and applying machine learning and deep learning methods for this process.
- Fine-tune an NLP solution based on the business problem and industry vertical.
- Evaluate various algorithms and approaches for the given task, dataset, and stage of the NLP product.
- Plan the lifecycle of the NLP product and produce software solutions following best practices around release, deployment, and DevOps for NLP systems.
- Understand best practices, opportunities, and the roadmap for NLP from a business and product leader's perspective.

You will also learn to adapt your solutions for different industry verticals like healthcare, finance, and retail. Moreover, you will learn about specific caveats you will encounter in each.

Structure of the Book

The book is divided into four sections. **Figure P-1** illustrates the chapter organization. Isolated chapters that are not directly connected to other chapters are easiest to skip while moving forward.

Chapter & Section Structure

Practical Natural Language Processing

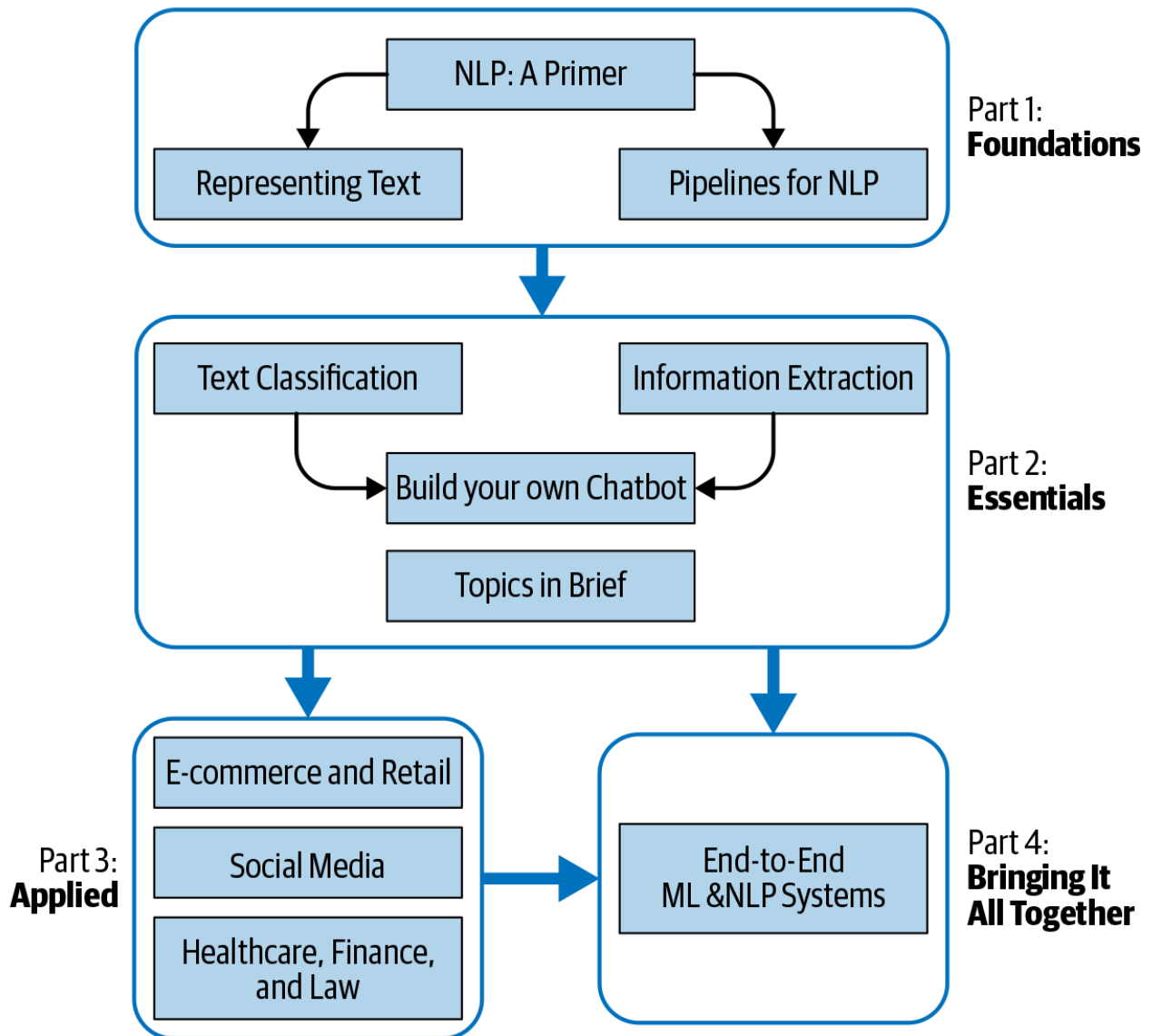


Figure P-1. How the book's sections are structured

Part I, *Foundations* acts as a bedrock for the rest of the book, by giving an overview of NLP (**Chapter 1**), discussing typical data processing and modeling pipelines used in building NLP systems (**Chapter 2**), and introducing different ways of representing textual data in NLP (**Chapter 3**).

Part II, *Essentials*, focuses on the most common NLP applications, with an emphasis on real-world use cases. Where possible, we show multiple solutions to the problem at hand to demonstrate how to choose among different options. Some applications include text classification (**Chapter 4**), information extraction (**Chapter 5**), and the building of chat bots (**Chapter 6**). We also introduce other applications, such as search, topic modeling, text summarization, and machine translation, along with a discussion about practical use cases (**Chapter 7**).

Part III, *Applied* (Chapters 8–10) specifically focuses on three industry verticals where NLP is heavily used, with a detailed discussion on those domains’ specific problems and how NLP is useful in addressing them.

Finally, **Part IV (Chapter 11)** brings all the learning together by dealing with the issues involved in end-to-end deployment of NLP systems in practice.

How to Read This Book

How one reads the book depends on their role and objective. For a data scientist or an engineer delving into NLP, we recommend reading Chapters 1–6 and then focusing on the particular domain or subproblem of interest. For someone in a leadership role, we recommend focusing on Chapters 1, 2, and 11. They might want to give extra focus to case studies for Chapters 3–7, which provide more ideas on the process of building NLP applications from scratch. A product leader might want to delve deep into the references provided for relevant chapters, as well as **Chapter 11**.

NLP applications for various domains can be different from the general problems covered in Chapters 3–7. That is why we have focused more on certain domains such as e-commerce, social media, healthcare, finance, and law. If your interest or work takes you to these areas, you can dig deeper into those chapters and corresponding references.

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file extensions.

Constant width

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

Constant width bold

Shows commands or other text that should be typed literally by the user.

Constant width italic

Shows text that should be replaced with user-supplied values or by values determined by context.

TIP

This element signifies a tip or suggestion.

NOTE

This element signifies a general note.

WARNING

This element indicates a warning or caution.

Using Code Examples

Supplemental material (code examples, exercises, etc.) is available for download at <https://oreil.ly/PracticalNLP>.

If you have a technical question or a problem using the code examples, please send email to bookquestions@oreilly.com.

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Further Information

In the **GitHub repo** you will find notebooks explaining various concepts covered in the book. The notebooks have been organized by chapter. We also provide additional notebooks, not necessarily covered in the book.

Book website: <http://www.practicalnlp.ai>

The world of NLP is always evolving. To stay updated on how concept mentioned in the book fit into the broader context one, two, and five years from now, follow our **blog**. We keep it updated with relevant writeups and articles, and tag every post with the book's corresponding chapter title.

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Part I. Foundations

Chapter 1. NLP: A Primer

*A language is not just words. It's a culture, a tradition,
a unification of a community,
a whole history that creates what a community is.
It's all embodied in a language.*

—Noam Chomsky

Imagine a hypothetical person, John Doe. He's the CTO of a fast-growing technology startup. On a busy day, John wakes up and has this conversation with his digital assistant:

John: "How is the weather today?"

Digital assistant: "It is 37 degrees centigrade outside with no rain today."

John: "What does my schedule look like?"

Digital assistant: "You have a strategy meeting at 4 p.m. and an all-hands at 5:30 p.m. Based on today's traffic situation, it is recommended you leave for the office by 8:15 a.m."

While he's getting dressed, John probes the assistant on his fashion choices:

John: "What should I wear today?"

Digital assistant: "White seems like a good choice."

You might have used smart assistants such as Amazon Alexa, Google Home, or Apple Siri to do similar things. We talk to these assistants not in a programming language, but in our natural language—the language we all communicate in. This natural language has been the primary medium of communication between humans since time immemorial. But computers can only process data in binary, i.e., 0s and 1s. While we can represent language data in binary, how do we make machines understand the language? This is where natural language processing (NLP) comes in. It is an area of computer science that deals with methods to analyze, model, and understand human language. Every intelligent application involving human language has some NLP behind it. In this book, we'll explain what NLP is as well as how to use NLP to build and scale intelligent applications. Due to the open-

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ended nature of NLP problems, there are dozens of alternative approaches one can take to solve a given problem. This book will help you navigate this maze of options and suggests how to choose the best option based on your problem.

This chapter aims to give a quick primer of what NLP is before we start delving deeper into how to implement NLP-based solutions for different application scenarios. We'll start with an overview of numerous applications of NLP in real-world scenarios, then cover the various tasks that form the basis of building different NLP applications. This will be followed by an understanding of language from an NLP perspective and of why NLP is difficult. After that, we'll give an overview of heuristics, machine learning, and deep learning, then introduce a few commonly used algorithms in NLP. This will be followed by a walkthrough of an NLP application. Finally, we'll conclude the chapter with an overview of the rest of the topics in the book. **Figure 1-1** shows a preview of the organization of the chapters in terms of various NLP tasks and applications.

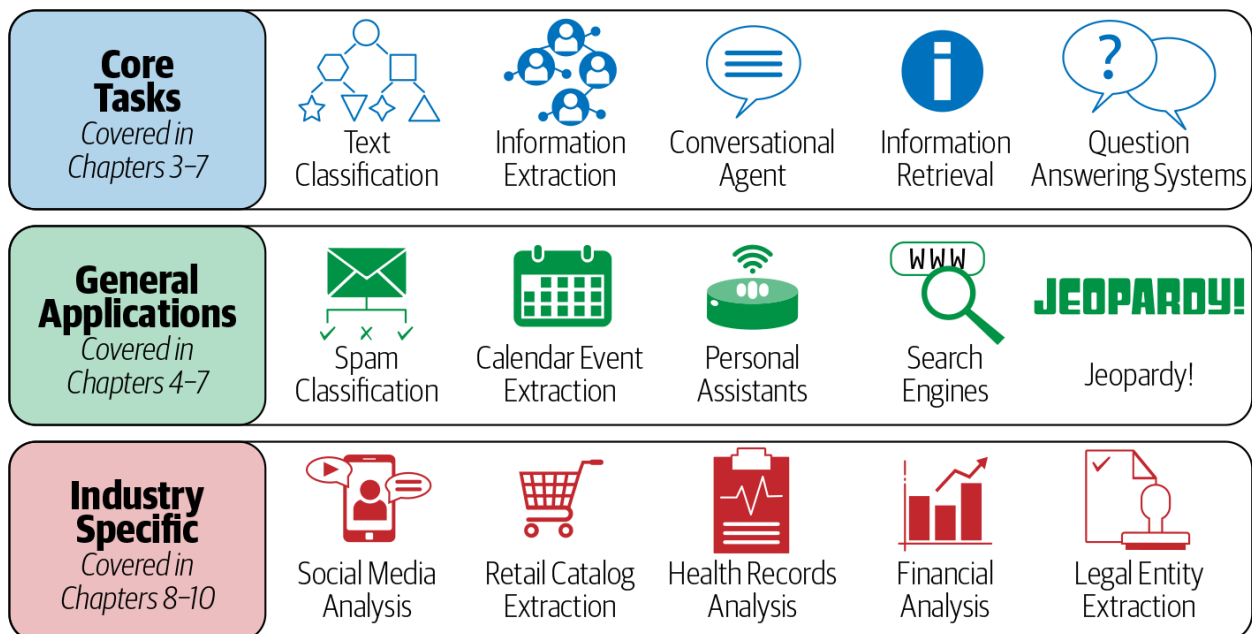


Figure 1-1. NLP tasks and applications

Let's start by taking a look at some popular applications you use in everyday life that have some form of NLP as a major component.

NLP in the Real World

NLP is an important component in a wide range of software applications that we use in our daily lives. In this section, we'll introduce some key applications and also take a look at some common tasks that you'll see across different NLP applications. This section reinforces the applications we showed you in **Figure 1-1**, which you'll see in more detail throughout the book.

Core applications:

- Email platforms, such as Gmail, Outlook, etc., use NLP extensively to provide a range of product features, such as spam classification, priority inbox, calendar event extraction, auto-complete, etc. We'll discuss some of these in detail in Chapters 4 and 5.
- Voice-based assistants, such as Apple Siri, Google Assistant, Microsoft Cortana, and Amazon Alexa rely on a range of NLP techniques to interact with the user, understand user commands, and respond accordingly. We'll cover key aspects of such systems in Chapter 6, where we discuss chatbots.
- Modern search engines, such as Google and Bing, which are the cornerstone of today's internet, use NLP heavily for various subtasks, such as query understanding, query expansion, question answering, information retrieval, and ranking and grouping of the results, to name a few. We'll discuss some of these subtasks in Chapter 7.
- Machine translation services, such as Google Translate, Bing Microsoft Translator, and Amazon Translate are increasingly used in today's world to solve a wide range of scenarios and business use cases. These services are direct applications of NLP. We'll touch on machine translation in Chapter 7.

Other applications:

- Organizations across verticals analyze their social media feeds to build a better and deeper understanding of the voice of their customers. We'll cover this in Chapter 8.
- NLP is widely used to solve diverse sets of use cases on e-commerce platforms like Amazon. These vary from extracting relevant information from product descriptions to understanding user reviews. Chapter 9 covers these in detail.
- Advances in NLP are being applied to solve use cases in domains such as healthcare, finance, and law. Chapter 10 addresses these.
- Companies such as Arria [1] are working to use NLP techniques to automatically generate reports for various domains, from weather forecasting to financial services.
- NLP forms the backbone of spelling- and grammar-correction tools, such as Grammarly and spell check in Microsoft Word and Google Docs.
- *Jeopardy!* is a popular quiz show on TV. In the show, contestants are presented with clues in the form of answers, and the contestants must phrase their responses in the form of questions. IBM built the Watson AI to compete with the show's top players. Watson won the first prize with a million dollars, more than the world champions. Watson AI was built using NLP techniques and is one of the examples of NLP bots winning a world competition.

- NLP is used in a range of learning and assessment tools and technologies, such as automated scoring in exams like the Graduate Record Examination (GRE), plagiarism detection (e.g., Turnitin), intelligent tutoring systems, and language learning apps like Duolingo.
- NLP is used to build large knowledge bases, such as the Google Knowledge Graph, which are useful in a range of applications like search and question answering.

This list is by no means exhaustive. NLP is increasingly being used across several other applications, and newer applications of NLP are coming up as we speak. Our main focus is to introduce you to the ideas behind building these applications. We do so by discussing different kinds of NLP problems and how to solve them. To get a perspective on what you are about to learn in this book, and to appreciate the nuances that go into building these NLP applications, let's take a look at some key NLP tasks that form the bedrock of many NLP applications and industry use cases.

NLP Tasks

There is a collection of fundamental tasks that appear frequently across various NLP projects. Owing to their repetitive and fundamental nature, these tasks have been studied extensively. Having a good grip on them will make you ready to build various NLP applications across verticals. (We also saw some of these tasks earlier in [Figure 1-1](#).) Let's briefly introduce them:

Language modeling

This is the task of predicting what the next word in a sentence will be based on the history of previous words. The goal of this task is to learn the probability of a sequence of words appearing in a given language. Language modeling is useful for building solutions for a wide variety of problems, such as speech recognition, optical character recognition, handwriting recognition, machine translation, and spelling correction.

Text classification

This is the task of bucketing the text into a known set of categories based on its content. Text classification is by far the most popular task in NLP and is used in a variety of tools, from email spam identification to sentiment analysis.

Information extraction

As the name indicates, this is the task of extracting relevant information from text, such as calendar events from emails or the names of people mentioned in a social media post.

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